



Linear equations in one unknown: Supported practice

AUTHORED practice

Subject: Mathematics

Topic: Linear equations

Concept: Operations that preserve the solution set

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Solve a one-unknown linear equation over the rationals

Question type: Constructed response

Provenance: Authored practice

Solve a one-unknown linear equation over the rationals

Solve $3x + 2 = 9$ over the rationals. Give the exact

1. The equation is a claim about x , not an instruction. What single

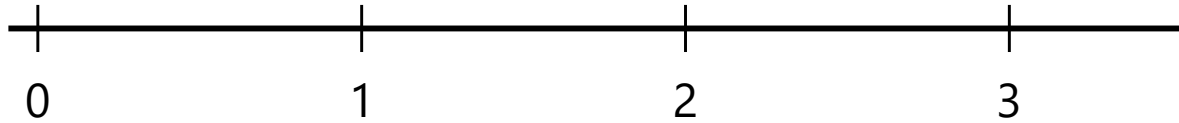
solution and verify it.

operation on both sides would
leave x with nothing added to it?

Conditions

- The unknown ranges over the rationals.
- Give the solution exactly, not as a rounded decimal.

Domain: the rationals · x (dimensionless)



Stage 1 of 2

Declared domain; Bracketing integers